Quaternion Volume: deterministic pilot observations

Observation-only status

This report records execution outputs; it makes no support, weakening, priority or scientific-status decision. Independent mathematical review remains separate. All instances are supplied synthetic lattices.

Trial and provenance

Experiment EXP-SSI-39bae9; hypothesis H-SSI-174d36; run RUN-SSI-5a435c. Code commit fa51ba977d0b57d45e0c65439ca9c67e1924e555. Seed 1729. Exact command: ["python3", "tools/quaternion_order_volume.py", "--seed", "1729", "--output-dir", "/workspace/lattice-quaternion/experiments/EXP-SSI-39bae9/runs/RUN-SSI-5a435c"]. One trial requested; completion/failure and resource observations are preserved in execution-receipt.json. The driver records full-tree dirtiness as unknown when not supplied; every source byte is admitted against its committed hash.

Recorded outputs

Exact fixture counters (integer counts, not independent statistical samples): {"all_exact_checks_passed": true, "candidate_elements": 81, "index_histogram": {"1": 2016, "12": 72, "15": 144, "19": 72, "2": 360, "20": 144, "3": 1152, "4": 360, "5": 720, "7": 144, "8": 72, "9": 360, "rank_deficient": 945}, "lattice_controls": 7, "ordered_pair_cases": 6561}. Supervisor receipt: {"artifact_sha256": {"check.stderr.log": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "check.stdout.log": "af8e9a371091dd2327af3784f48f79d137de7e7e00403d323475183ee8867f36", "command.txt": "2be31d1fd9c7bcc1b6673f7210653e9037364dff7cc83fcdc0950de2d68cc872", "environment.json": "4a904119e58348e412fae8b2d72dac0c06c943084a1d6c3e0fee6b13907438dd", "launch.json": "a4f8e996c6fd5d0056fafadaa0d8da9230c2174387120ba65ebcbd959b9ad643", "manifest.yaml": "f4b2b6fc15bd7b1e643c9dda489802f5f64c01e4313e043284fe5ff129e5aa40", "raw-result.json": "6ea07eb0c27181e2f7652ee8d90bb1621219caceacf1828d1e8a0c2f1297015f", "stderr.log": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855", "stdout.log": "2446de836ce4a897fa690c0007545d670adbfca15b0a4f4ea980549e21d91007"}, "authority": {"commit": "fa51ba977d0b57d45e0c65439ca9c67e1924e555", "dispatch_plan_sha256": "41390b103bece15c82b48b65ad0ff381190d0099485906cff8739572ae398b2b", "epoch": 1, "expires_at": "2026-10-14T16:13:17+00:00", "owner": "lattice-executor-20261014"}, "check_returncode": 0, "detail": null, "finished_at": "2026-10-14T15:16:46.691305+00:00", "plan_sha256": "b6065d3e3104e4973b6e3b0bc67b9012032b7dcd2243f7a82b9427fbcb436a22", "published": false, "returncode": 0, "run_id": "RUN-SSI-5a435c", "schema": "crypto.autoresearch.trial_receipt.v1", "scientific_conclusion": null, "started_at": "2026-10-14T15:16:41.866215+00:00", "status": "output_validated", "trial_id": "exact-pilot-seed1729"}

Control scope and graph

Four retained lattice controls. Equal additive index does not certify multiplicative closure. All plotted values come directly from raw-result.json; no new sampling is performed. Uncertainty bars are inapplicable to these deterministic exact values. Plot source and extracted data are retained.

Limitations and missing data

One supplied maximal quaternion order at p=2, not arbitrary supersingular curve maps. A small Gram determinant measures additive index, not multiplicative closure or subgroup action. Noncommuting generators can still give the proper Lipschitz suborder. Enumeration and scoring exclude isogeny discovery/evaluation and establish no performance advantage. Characteristic-prime division and finite-field realization remain separate unimplemented obligations. The integrity checker replays the implementation; it is not independent research review. No ECDLP canonical graph or frontier was changed.

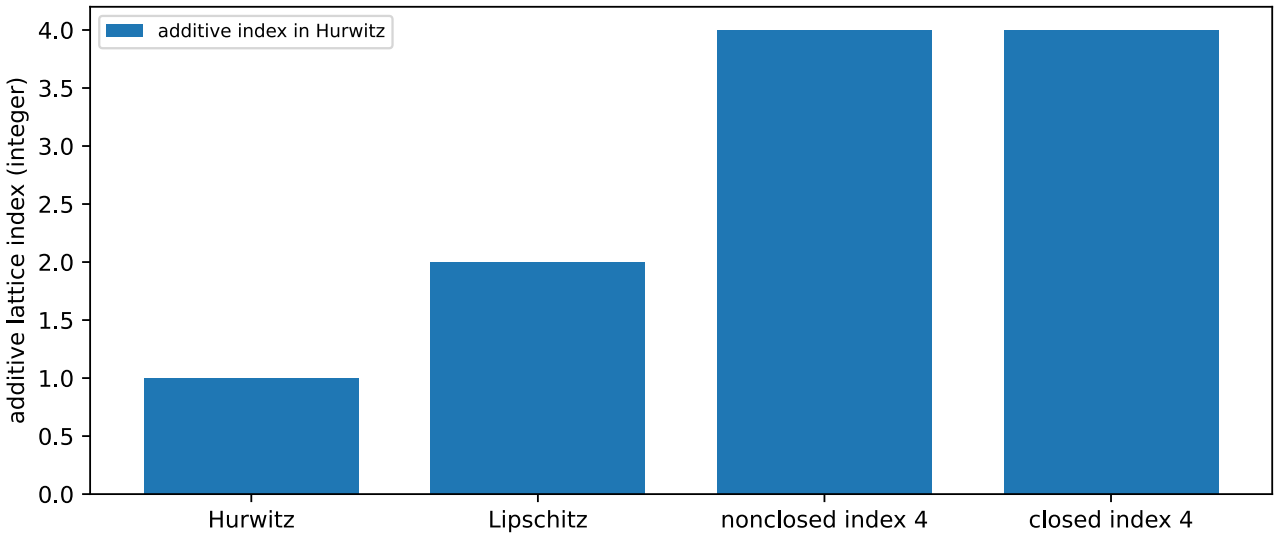

Figure 1: Four retained lattice controls. Equal additive index does not certify multiplicative closure. Run RUN-SSI-5a435c.